University of Michigan

NERS 515

Nuclear Measurements Laboratory

Laboratory 5: Proportional Counters

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Abstract

A windowless 2π gas-flow proportional counter filled with P-10 gas was characterized using Am-241, Cl-36 and Co-60 sources. A shaping-time scan of the Am-241 alpha peak gave a ballistic deficit of $28.85 \pm 0.05\%$ at 0.5 μs relative to 10 μs, and 3 μs was chosen as the shortest shaping time without a significant deficit. Counting curves for Am-241 alphas and Cl-36 betas both rose to short, sloped plateaus, of $8.4 \pm 0.2\%$ and $8.5 \pm 0.4\%$ per 100 V, and the beta pulses were much smaller than the alpha pulses, as expected. The alpha peak gave an energy calibration factor of 2.317 ± 0.001 keV/channel at 800 V. The Cl-36 source gave an absolute efficiency of 0.6234 ± 0.0005 and an intrinsic efficiency of 1.247 ± 0.001, above the expected limits of 0.5 and 100%, most likely because of backscattering from the source backing. The response to Co-60 gamma rays alone was weak, peaking at 1712 ± 8 s^{-1}. Both the alpha and gamma count rates dropped at the highest voltages, where the amplifier output saturated.

1 Introduction

This experiment characterizes a windowless 2π gas-flow proportional counter filled with P-10 gas (90% Ar, 10% CH_4). The purpose was to establish how its output depends on applied voltage, amplifier settings, and radiation type, and to find a stable operating point for counting [1].

Proportional counters are detectors that combine the characteristics of ion chambers and Geiger-Müller (GM) tubes. They use gas multiplication to amplify the charge from each event by a factor of hundreds to thousands, like GM tubes, while keeping the pulse amplitude proportional to the energy deposited in the gas, similar to ion chambers [2]. Therefore, alpha particles and beta particles produce pulses of different sizes, making counting of each of these particles visually different and easily differentiated.

Four sets of measurements were performed. First, the amplifier shaping time, gain, and lower-level discriminator were optimized using an ^{241}Am alpha source. Counting curves were then recorded for alpha and beta sources, and the alpha spectrum was energy calibrated. Next, the absolute counting efficiency was determined with a calibrated Cl-36 source. Finally, the response of the counter to gamma rays alone was examined. The underlying theory and the detailed procedure are given in Sections 2 and 3.2.

2 Theory

2.1 Ionization in the fill gas

A charged particle slowing down in the gas loses energy by exciting and ionizing gas molecules. The average energy spent per ion pair, the W-value, is much larger than the ionization potential, because part of the energy goes into excitation. According to Table 6.2 from Knoll [2], that value for P-10 gas, which was used in our proportional counter, is $W = 26$ eV/ion-pair.

The mean number of primary ion pairs is therefore

$$n_0 = \frac{E}{W} \tag{1}$$

where E is the energy deposited in the gas.

2.2 Gas Multiplication

Close to the thin anode wire, the electric field is strong enough for an electron to ionize further gas molecules between collisions. Each freed electron does the same, producing a cascade called a Townsend avalanche [2]. Nearly all primary electrons are formed outside this small region and drift into it. Each is therefore multiplied by the same average factor M, and the collected charge is

$$Q = n_0 e M \tag{2}$$

where Q is the charge collected. So, Q, and therefore the pulse amplitude, stays proportional to the deposited energy E. The Diethorn expression relates M to the applied voltage, as shown in

$$\ln M = \frac{V}{\ln(b/a)} \frac{\ln 2}{\Delta V} \left[\ln \frac{V}{pa \ln(b/a)} - \ln K \right] \tag{3}$$

For P-10, $K = 4.8 \times 10^4$ V/(cm·atm) and $\Delta V = 23.6$ V, according to Table 6.1 from [2]. The logarithmic term varies slowly, so M grows nearly exponentially with V. Each equal voltage step multiplies the pulse amplitude by about the same factor, so ln(amplitude) vs. V is a straight line (Fig. 6.11 of [2]). Table 6.3 of Knoll shows how sensitive the gain is in a P-10 counter. A 1% rise in voltage increases M by about 17%, roughly a factor of 2–3 per 100 V. A 1% rise in gas pressure lowers M by about 9%. The exact numbers depend on the counter dimensions, so they are only a rough guide for our counter, but small drifts in voltage or gas pressure can visibly change the gain [2].

2.3 Pulse formation, shaping time and ballistic deficit

Almost all of the signal comes from the positive ions formed in the avalanche moving away from the anode wire, not from the electrons, so the pulse reaches its full value only when the ions arrive at the cathode, after the ion collection time shown as

$$t^+ = \frac{(b^2 - a^2) p \ln(b/a)}{2\mu V_0} \tag{4}$$

Typical values give several hundred microseconds. Most of the amplitude, however, comes from the early part of the ion drift near the wire. Half of the full amplitude is reached at

$$t_{1/2} = \frac{a}{a+b} t^+ \tag{5}$$

For typical dimensions, $t_{1/2}$ is only about 0.25% of t^+, a fraction of a microsecond [2]. A shaping time of a few microseconds therefore cuts off the slow remainder of the pulse, and the resulting loss in amplitude is the ballistic deficit [2].

If every pulse had the same shape, the deficit would only scale all pulses down by the same factor. In a proportional counter, ionization starts at different radii, so rise times vary. Therefore the deficit varies from pulse to pulse [2]. Shaping times of several microseconds avoid this, but longer shaping adds pile-up and noise. The best choice is the shortest shaping time at which the peak centroid stops increasing [2]. The ballistic deficit, based on Knoll's definition, is estimated by comparing the centroids at a short and a long shaping time[1]:

$$BD(\%) = \frac{\tilde{C}_{10\mu s} - \tilde{C}_{0.5\mu s}}{\tilde{C}_{10\mu s}} \times 100 \tag{6}$$

2.4 Pulse height spectrum and energy calibration

Monoenergetic alpha particles stopped in the fill gas give a single peak in the differential pulse height spectrum [2]. Am-241 emits alphas at 5.486 MeV (85%) and 5.443 MeV (13%) [2]. The lines are only 0.8% apart, much less than the few-percent resolution of a proportional counter, so they appear as one peak. Below the peak, there is a tail of smaller pulses from alphas that lose energy in the source or its own cover, or alphas that reach the chamber wall before stopping. Noise and single-ion-pair events sit at the lowest amplitudes [2].

The electronics also limit what is recorded. The amplifier output is linear only up to about 10 V, and the MCA bins pulses only up to its last channel. When the gas gain pushes pulses past these limits, they are not recorded. A centroid near full scale is then unreliable, and the counts in an ROI can drop even though more pulses exceed the LLD.

For a linear system, pulse height channel and deposited energy are related by

$$E = k\,(\mathrm{ch} - \mathrm{ch}_0) \tag{7}$$

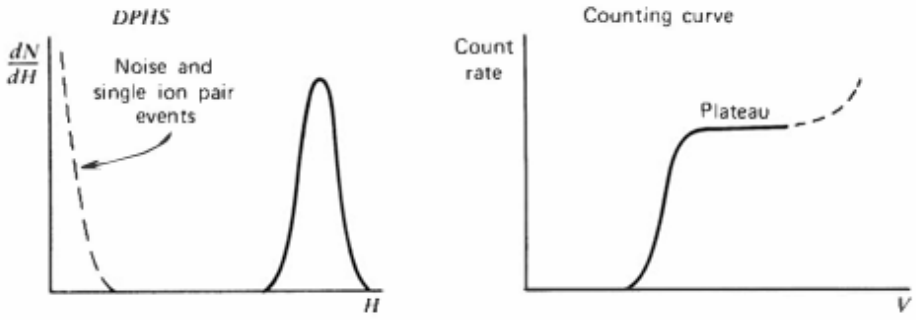
where ch_0 is the channel of zero amplitude, either measured (for example with a pulser) or taken as 0. The factor k comes from a peak of known energy and holds only for the voltage, amplifier gain and shaping time at which it was measured.

2.5 Counting Curves

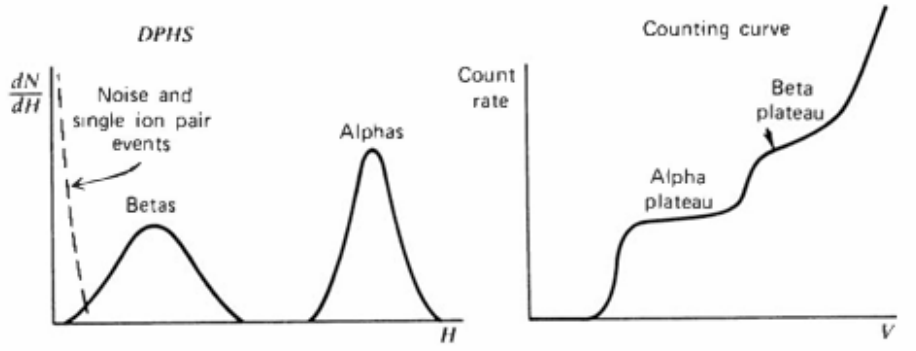
A counting curve is the count rate above a fixed LLD, recorded as the voltage (and so the gain) is raised [2]. It mirrors the integral pulse height distribution. At low voltage only the largest pulses cross the LLD, and smaller ones are added as the voltage rises. The slope dr/dV therefore maps the differential distribution, with higher voltage corresponding to smaller pulses. Where the LLD sits in a valley of the differential spectrum, the curve flattens into a plateau, and the operating point is chosen there. Plateau flatness is usually quoted as the percent change in count rate per 100 V [2].

[1] Ballistic deficit for the specified times in [1].

Alpha particles stop in the gas and give one well-separated peak, so the alpha curve rises sharply through a knee and then flattens into a wide plateau. A beta's range is much longer than the chamber, so it deposits only a small part of its energy in the gas. Beta pulses are therefore much smaller than alpha pulses and spread over a wide range that merges into the noise. The beta curve rises more gradually and has a shorter, steeper plateau. A source emitting both gives two plateaus, first alpha and then alpha and beta together (Fig. 1). At high gain the plateau ends when spurious afterpulses start to be counted and the rate climbs [2].



(a) Alpha source: one well-separated peak and a wide plateau.



(b) Mixed alpha and beta source: a second, shorter and steeper plateau.

Figure 1: Differential pulse height spectra (left) and the corresponding counting curves (right). Adapted from [2].

2.6 Detection Efficiency

The absolute efficiency compares counts with the particles the source emits, and the intrinsic efficiency compares them with the particles that enter the detector [2]:

$$\varepsilon_{\text{abs}} = \frac{\text{number of pulses recorded}}{\text{number of radiation quanta emitted by source}} \tag{8}$$

$$\varepsilon_{\text{int}} = \frac{\text{number of pulses recorded}}{\text{number of radiation quanta incident on detector}} = \varepsilon_{\text{abs}} \frac{4\pi}{\Omega} \tag{9}$$

In our internal-source counter the source sits at the base of the counting volume, so $\Omega \approx 2\pi$. Every alpha or beta that enters the gas produces a pulse, so on the plateau $\varepsilon_{\text{int}} \approx 100\%$ and $\varepsilon_{\text{abs}} \approx 50\%$ [2].

Backscattering can raise the measured rate for beta sources [2]. Electrons emitted into a thick source backing can scatter back out through the source surface, increasing the apparent yield toward the detector. The effect is strongest for low electron energies and high-Z backings. Below about 1 MeV, the backscattered fraction is about 0.1 to 0.15 for aluminum and 0.3 for copper (Fig. 2.17 of [2]).

The source activity on the measurement date follows from its marked activity A_0:

$$A(t) = A_0\, e^{-\lambda t}, \qquad \lambda = \frac{\ln 2}{T_{1/2}} \tag{10}$$

For Cl-36, $T_{1/2} = 3.08 \times 10^5$ y [2], so decay over a few months is negligible.

2.7 Gamma Response

Gamma rays interact very weakly in the counter gas, so most of the response comes from electrons knocked out of the counter walls that reach the gas [2]. These electrons lose varying amounts of energy before reaching the gas, so their pulses are small and spread over a wide range with no peak, and the gamma counting curve should resemble a beta curve at a much lower rate.

2.8 Counting Statistics

A recorded number of counts N has a standard deviation $\sigma_N = \sqrt{N}$ [2]. The counting time is known much more precisely, so a count rate has the same fractional error as the counts [2]:

$$r = \frac{N}{t}, \qquad \sigma_r = \frac{\sqrt{N}}{t} \tag{11}$$

For a quantity $u(x, y, \ldots)$ calculated from independent measured values, the error propagation formula gives [2]

$$\sigma_u^2 = \left(\frac{\partial u}{\partial x}\right)^2 \sigma_x^2 + \left(\frac{\partial u}{\partial y}\right)^2 \sigma_y^2 + \cdots \tag{12}$$

For a sum or difference this reduces to the quadrature sum, $\sigma_u = \sqrt{\sigma_x^2 + \sigma_y^2}$.

A Gaussian peak has standard deviation $\sigma = \text{FWHM}/2.35$ [2]. Just as the error of a mean of N measurements is $\sigma/\sqrt{N}$ [2], the statistical uncertainty of the centroid of a peak containing A counts is

$$\sigma_C = \frac{\text{FWHM}/2.35}{\sqrt{A}} \tag{13}$$

3 Equipment and Procedure

3.1 Equipment

The details of all equipment used in this lab are listed in Table 1. In addition, we used a proportional counter connected to a P-10 gas line, a preamplifier, and Co-60, Am-241 and Cl-36 sources as specified in the lab manual [1]. Figure 2 shows the connections.

Purpose	**Manufacturer**	**Model**
Oscilloscope	Agilent Technologies	DSO6052A
High Voltage Power Supply	Ortec	556
Amplifier	Ortec	572A
Mulit-Channel Analyzer	Ortec	Easy-MCA

Table 1: List of equipment used in Laboratory 5.

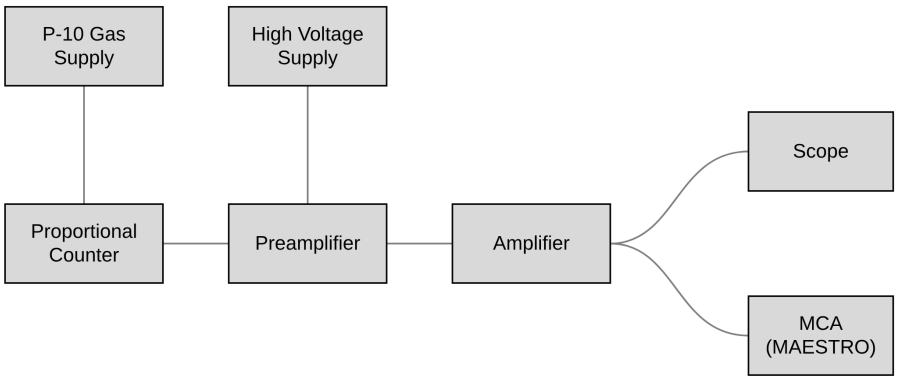


Figure 2: Set-up diagram.

3.2 Experimental Procedure

3.2.1 Shaping time

1. With Am-241 in the chamber at 800 V, the amplifier gain was set to place the alpha peak on scale and was kept fixed for the rest of the lab.

2. A $\sim$1 min spectrum was recorded at each available shaping time, and the centroid and gross area of an ROI around the alpha peak were noted.

3. The shortest shaping time without a significant ballistic deficit was chosen and used for the rest of the lab.

3.2.2 LLD selection

1. For each counting curve, an LLD channel was chosen so that no counts were recorded at the lowest voltage of the scan.

2. Counts above the LLD were taken as the sum of an ROI from the LLD channel to the last MCA channel.

3.2.3 Alpha counting curve

1. With Am-241 in the chamber, the high voltage was raised in steps, with smaller steps through the knee, waiting ∼30 s after each change.
2. At each voltage, the counts above the LLD and the peak centroid were recorded for a fixed live time, while the amplifier output was watched on the oscilloscope for saturation.
3. The voltage was raised until the count rate leveled off.

3.2.4 Beta counting curve

1. The Am-241 source was replaced with the Cl-36 beta source.
2. The counts above the LLD were recorded for a fixed live time at each voltage step, without exceeding 1700 V.

3.2.5 Absolute efficiency

1. An operating voltage was chosen from the beta counting curve, with the same LLD and gain.
2. The Cl-36 source was counted for 1 min.
3. The marked activity was decay-corrected to the measurement date and compared with the measured count rate.

3.2.6 Gamma response

1. The Co-60 source was placed upside down in the chamber so that its backing stopped the betas.
2. The counts above the LLD were recorded for a fixed live time at each voltage step, without exceeding 1700 V.

4 Data

Count uncertainties follow Eq. 11. All count times are Maestro live time, and we did not take a background count. The amplifier gain was 50 (coarse 100, fine 0.5) for the whole lab and the shaping time was 3 μs outside the shaping-time scan.

We convert MCA channels to pulse amplitude with the calibration from our Lab 2 report, channel $= 765.8\,V + 27.6$, so

$$V = \frac{\text{channel} - 27.6}{765.8}, \tag{14}$$

with an amplitude uncertainty of 25.4 mV [3].

4.1 Shaping time

Table 2 gives the Am-241 alpha peak centroid and gross area for each shaping time at 800 V, about 1 minute each. The centroid rises from 1723 at 0.5 μs to 2422 at 10 μs.

Shaping time (μs)	Centroid (channel)	Gross area (counts)
0.5	1723.08	599257 ± 774
1	2025.16	624639 ± 790
2	2215.87	632332 ± 795
3	2367.29	712298 ± 844
6	2382.50	738092 ± 859
10	2421.64	742365 ± 861

Table 2: Alpha peak centroid and gross area vs shaping time, Am-241 at 800 V.

4.2 Alpha counting curve

Table 3 lists the gross counts above LLD channel 1738 (2.23 V) and the peak centroid, 30 s per point. We kept raising the voltage past the suggested 300–800 V [1] until the count rate leveled off. Counts first appeared at 400 V. At 1050 V the counts dropped from 669,674 to 286,105. We re-measured that point several times and got the same result. From 500 to 850 V the recorded centroid stays between 1744 and 2133, close to the LLD, and does not rise steadily.

HV (V)	Counts (30 s)	Centroid	HV (V)	Counts (30 s)	Centroid
300	0 ± 0	–	725	121059 ± 348	1753.07
400	23 ± 5	–	750	224018 ± 473	1881.92
500	398 ± 20	1777.10	775	294972 ± 543	2118.21
550	2516 ± 50	1764.00	800	390186 ± 625	1743.68
600	11163 ± 106	1760.00	850	563218 ± 750	2133.17
625	19972 ± 141	1765.85	900	617516 ± 786	2738.47
650	32190 ± 179	1754.00	1000	669674 ± 818	3232.47
675	46571 ± 216	1822.40	1050	286105 ± 535	7166.13
700	61980 ± 249	1784.90			

Table 3: Alpha counting curve, Am-241: gross counts above LLD channel 1738 (2.23 V) and peak centroid channel.

4.3 Beta counting curve

Table 4 lists the Cl-36 counts above LLD channel 307 (0.36 V), 30 s per point. Counts first appeared at 1100 V and we stopped at 1550 V. The centroid stays near the LLD (305–335) up to 1450 V, then reads 508 and 773.

4.4 Absolute efficiency

We counted the Cl-36 source for 60 s at 1500 V, picked from the beta counting curve, with the same LLD (channel 307, 0.36 V) and gain: 1,428,191 ± 1195 counts. The source was marked 1.032 μCi on August 1, 2026.

HV (V)	Counts (30 s)	Centroid
1000	0 ± 0	–
1100	23161 ± 152	328.36
1200	97809 ± 313	334.88
1300	242325 ± 492	304.65
1400	345372 ± 588	311.34
1450	521827 ± 722	320.22
1500	664725 ± 815	507.70
1550	692883 ± 832	773.28

Table 4: Beta counting curve, Cl-36: gross counts above LLD channel 307 (0.36 V) and centroid channel.

HV (V)	Counts (30 s)
1000	0 ± 0
1200	10490 ± 102
1300	24776 ± 157
1400	43711 ± 209
1500	51324 ± 227
1550	50648 ± 225
1600	51362 ± 227
1650	48634 ± 221
1700	43041 ± 207

Table 5: Gamma response, Co-60 turned upside down: gross counts above LLD channel 229 (0.26 V).

4.5 Gamma response

Table 5 lists the counts with the Co-60 source turned upside down to block the betas, LLD channel 229 (0.26 V), 30 s per point. The counts rise to about 51,000 at 1500–1600 V and then fall at 1650 and 1700 V. At the same time, we observed some saturation in the amplifier output on the scope.

5 Analysis of Data

A rate from N counts in time t is $r = N/t$ with $\sigma_r = \sqrt{N}/t$ (Eq. 11). Other uncertainties are propagated with Eq. 12.

5.1 Shaping time and ballistic deficit

We compute the ballistic deficit with Eq. 6.

The centroid uncertainty is given by Eq. 13. We did not record the FWHM, so we cannot evaluate this. With $A \approx 6$–7.4×10^5 counts, even a peak several hundred channels wide gives σ_c below one channel, so we use $\sigma_c = 1$ channel as an upper bound. This only covers counting statistics, not the ROI placement or gain drift (Section 5.3).

Ballistic deficit at 0.5 μs ($C_{0.5} = 1723.08$, $C_{10} = 2421.64$, $\sigma_c = 1$)

$$\mathrm{BD} = \frac{2421.64 - 1723.08}{2421.64} \times 100\% = 28.85\%$$

$$\sigma_{\mathrm{BD}} = 100\%\,\sigma_c \sqrt{\left(\frac{1}{C_{10}}\right)^2 + \left(\frac{C_{0.5}}{C_{10}^2}\right)^2} = 100\% \sqrt{(4.13 \times 10^{-4})^2 + (2.94 \times 10^{-4})^2} = 0.05\%$$

The ballistic deficit at 0.5 μs relative to 10 μs is $28.85 \pm 0.05\%$. Figure 3 shows the centroid climbing steeply up to 3 μs and then leveling off: the 3 and 6 μs centroids differ by only 15 channels. We chose 3 μs as the shortest shaping time without a significant deficit.

This matches the expected behavior (Section 2.3). We did not measure the rise time of the

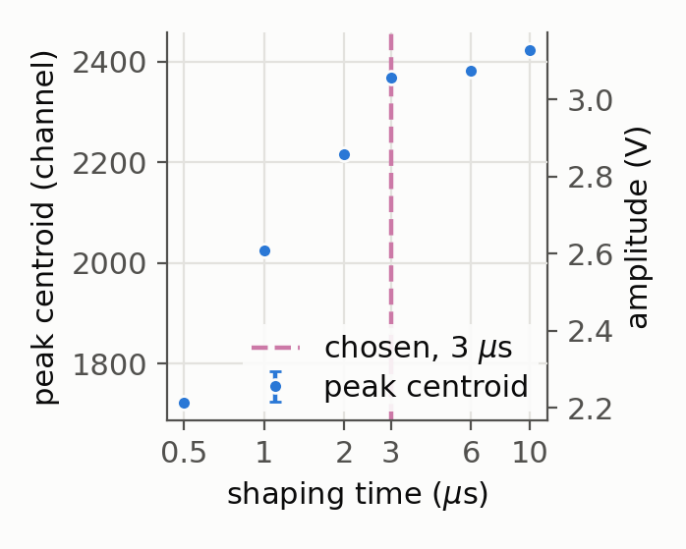


Figure 3: Alpha peak centroid vs shaping time at 800 V. Right axis: amplitude from the Lab 2 calibration (Eq. 14). Dashed line: chosen 3 μs. Error bars (± 1 channel, upper bound) are smaller than the markers.

amplifier output on the scope. A longer shaping time gives an output pulse that rises more slowly and peaks later, with an amplitude closer to the full value [2].

5.2 Alpha counting curve

All rates are in Appendix A and Figure 4a shows the counting curve.

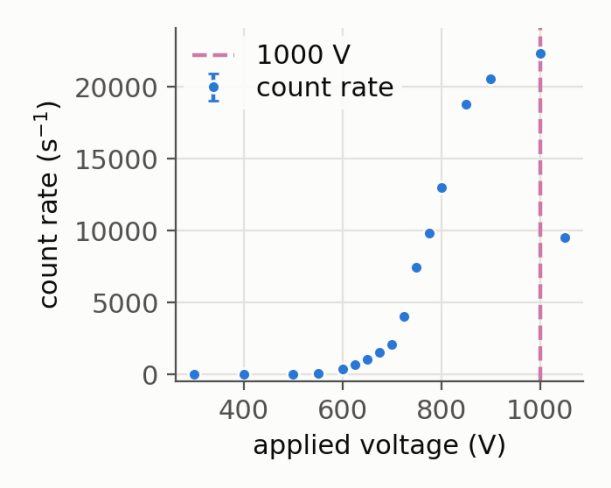
Rate at 1000 V and plateau slope over 900–1000 V

$$r_{1000} = 669674/(30\text{ s}) = 22322\text{ s}^{-1}, \qquad \sigma = \sqrt{669674}/(30\text{ s}) = 27\text{ s}^{-1}$$
$$r_{900} = 617516/(30\text{ s}) = 20584\text{ s}^{-1}, \qquad \sigma = \sqrt{617516}/(30\text{ s}) = 26\text{ s}^{-1}$$
$$s_{\%} = 100\,\frac{r_{1000} - r_{900}}{r_{900}} = 100\,\frac{1738.6}{20584} = 8.4\%\text{ per 100 V},$$
$$\sigma_{s_{\%}} = 100\,\frac{\sqrt{26^2 + 27^2}}{20584} = 0.2\%\text{ per 100 V}$$

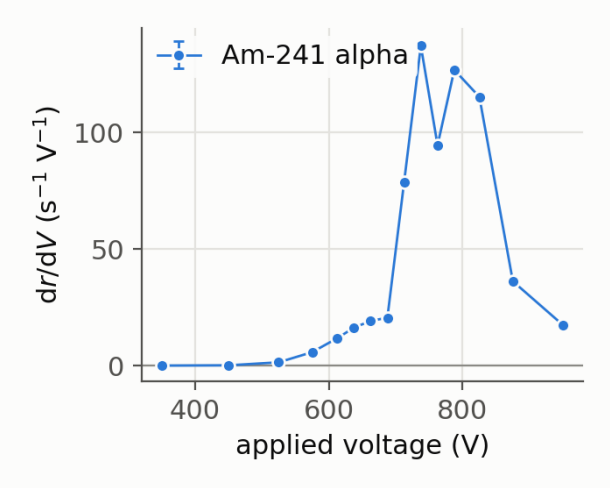
The expected shape is described in Section 2.5. Ours has the rise, from 400 to 850 V, but flattens only over 900–1000 V, with a slope of $8.4 \pm 0.2\%$ per 100 V. So the plateau is barely established and not flat.

The drop to 9537 s^{-1} at 1050 V is not expected (Section 2.5). The cause was likely due to saturation since the drop appears exactly where the peak reaches about 9 V (Section 2.4). We left 1050 V out of the plateau and took 1000 V as its start. No later section uses the alpha operating point, so the drop does not affect the other results.

Differential counting curve. Following Section 2.5, we take the slope between neighboring points, $dr/dV \approx (r_2 - r_1)/(V_2 - V_1)$, at the midpoint voltage, leaving out 1050 V.



(a) Counting curve



(b) Differential counting curve

Figure 4: Am-241 alphas, LLD channel 1738 (2.23 V). (a) Count rate vs voltage; dashed line at 1000 V, start of the plateau. (b) dr/dV between neighbouring points of (a), plotted at the midpoint voltage; higher voltage corresponds to smaller pulses. Error bars ($\sqrt{N}/t$ and Eq. 12) are smaller than the markers.

Slope between 725 and 750 V

$$\frac{dr}{dV} = \frac{7467.3 - 4035.3}{25\ \text{V}} = 137.3\ \text{s}^{-1}\text{V}^{-1}, \qquad \sigma = \frac{\sqrt{15.8^2 + 11.6^2}}{25\ \text{V}} = 0.8\ \text{s}^{-1}\text{V}^{-1}$$

Figure 4b has three features. The main one is a single peak between 700 and 850 V, with its maximum near 740 V. This is the alpha peak sweeping past the LLD, and it holds most of the counts. The peak has a dip at 760 V that is far larger than its error bar. A small gain change between neighbouring points (Section 5.3) is the likely cause. Before it, between 600 and 700 V, there is a smaller shoulder of about 20 $\text{s}^{-1}\ \text{V}^{-1}$ that contains roughly 1700 s^{-1}, about 8% of the plateau rate. These are pulses larger than the main peak and we cannot tell their origin without the spectrum. After the peak, the slope falls because of a tail of small pulses that only cross the LLD at the highest gains.

5.3 Alpha pulse amplitude vs voltage

Section 2.2 predicts a straight line on a log axis, with a factor of 2–3 per 100 V. We compare neighbouring points with the two-point slope $s = \ln(C_2/C_1)/\Delta V$.

Two-point slope, 900–1000 V ($C_1 = 2738.47$, $C_2 = 3232.47$, $\sigma_c = 1$)

$$s = \frac{\ln(3232.47/2738.47)}{100\ \text{V}} = 1.66 \times 10^{-3}\ \text{V}^{-1},$$
$$\sigma_s = \frac{\sqrt{(1/2738.47)^2 + (1/3232.47)^2}}{100\ \text{V}} = 5 \times 10^{-6}\ \text{V}^{-1}$$

The same for 1000–1050 V gives $s = (1.592 \pm 0.001) \times 10^{-2}\ \text{V}^{-1}$.

From 500 to 850 V the centroid stays within a few hundred channels of the LLD (hollow

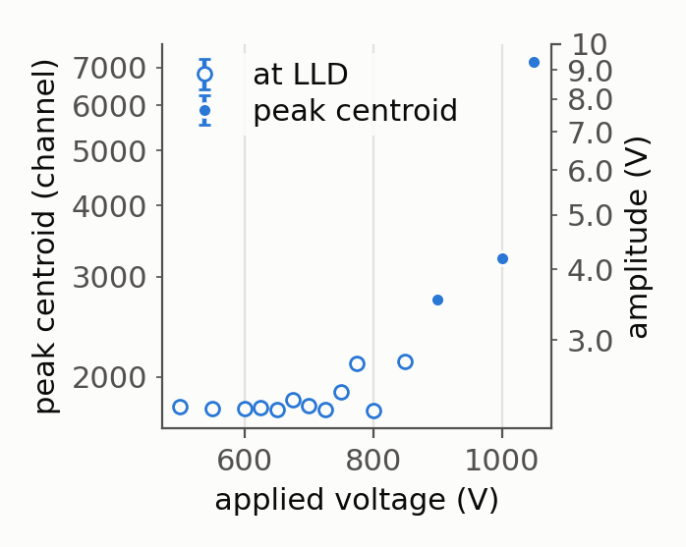


Figure 5: Alpha peak centroid vs voltage on a log axis. Right axis: amplitude from the Lab 2 calibration (Eq. 14). Hollow: 500–850 V, where the centroid sits at the LLD. Error bars (±1 channel) are smaller than the markers.

points in Figure 5). In this range the alpha peak was either below the LLD with only its upper edge counted, or not yet separated from the noise. Either way the number is the centroid of the counts above the LLD, not of the peak, so we leave these points out. This does not affect the counting curve, which only uses counts above the LLD and that range is where the alpha pulses cross the LLD.

That leaves 900, 1000 and 1050 V, and the 1050 V point is likely affected by saturation (previous section). The two slopes differ by about a factor of ten, so we cannot confirm the expected straight-line trend (Section 2.2) from our data.

The gas gain could have changed between sections. In the shaping-time scan the peak sat at channel 2367 (3.06 V) at 800 V and 3 μs, but on the counting curve at the same voltage, amplifier gain and shaping time the centroid was at the LLD (2.23 V), and the 800 V count rate was only about 60% of the plateau rate. Gas multiplication is very sensitive to small changes in high voltage and gas pressure (Section 2.2). We did not check a reference peak at the start of each section, so we cannot tell which one changed.

5.4 Energy calibration

We calibrate from the alpha peak centroid but without the .SPE files we cannot show the calibrated spectrum. The two Am-241 lines are not resolved (Section 2.4), so we assign 5.486 MeV to the peak. We use Eq. 7 with $\mathrm{ch}_0 = 0$. We use the 3 μs shaping-time run, where the ROI was drawn around the whole peak.

Calibration factor (800 V, gain 50, 3 μs, $C = 2367.29$)

$$k = \frac{5486\ \text{keV}}{2367.29} = 2.317\ \text{keV/channel}, \qquad \sigma_k = k\,\frac{\sigma_c}{C} = 2.317\,\frac{1}{2367.29} = 0.001\ \text{keV/channel}$$

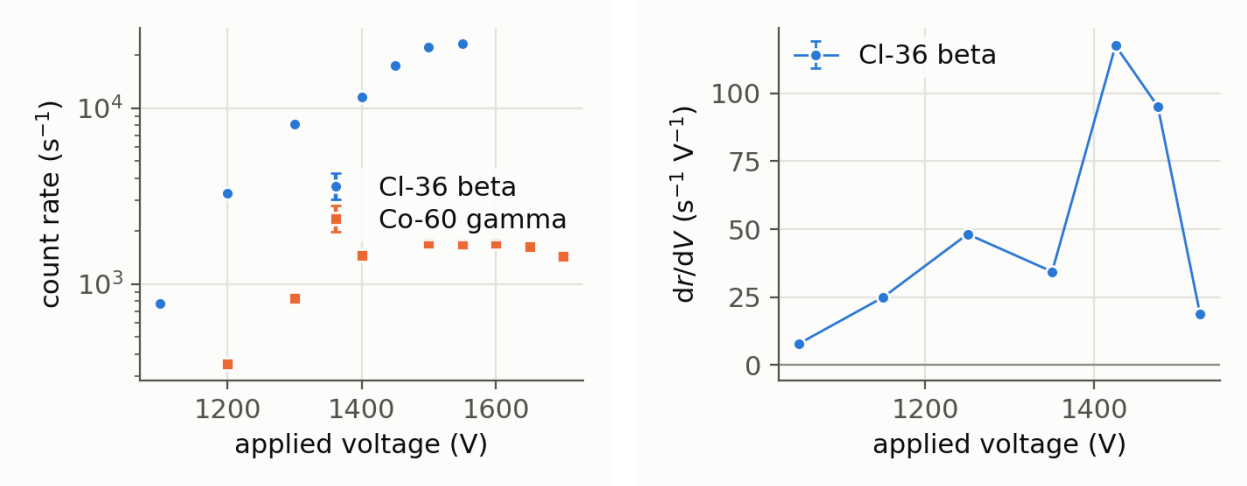
This factor holds only at 800 V, gain 50, 3 μs shaping, and the gas gain during the shaping-time runs. The same calculation at 1000 V on the counting curve would give 1.70 keV/channel. The differential counting curve (Figure 4b) gives the shape of the pulse height distribution. It agrees with the expected one (Section 2.4) apart from the small high-amplitude shoulder and the low-amplitude tail described above.

5.5 Beta counting curve

Figure 6a shows the Cl-36 beta counting curve with the gamma curve; the rates are in Appendix A.

Rate at 1500 V and plateau slope over 1500–1550 V

$$r_{1500} = 664725/(30\ \text{s}) = 22158\ \text{s}^{-1}, \qquad \sigma = 27\ \text{s}^{-1}$$
$$r_{1550} = 692883/(30\ \text{s}) = 23096\ \text{s}^{-1}, \qquad \sigma = 28\ \text{s}^{-1}$$
$$s_{\%} = 100\,\frac{r_{1550} - r_{1500}}{r_{1500}} \cdot \frac{100\ \text{V}}{50\ \text{V}} = 8.5\%\ \text{per 100 V},$$
$$\sigma_{s_{\%}} = 100\,\frac{\sqrt{27^2 + 28^2}}{22158} \cdot 2 = 0.4\%\ \text{per 100 V}$$



(a) Counting curves

(b) Beta differential counting curve

Figure 6: (a) Cl-36 betas (LLD channel 307, 0.36 V) and Co-60 gammas (LLD channel 229, 0.26 V). (b) Cl-36 dr/dV between neighbouring points of (a), plotted at the midpoint voltage. Error bars ($\sqrt{N}/t$ and Eq. 12) are smaller than the markers.

Section 2.5 predicts a gradual rise and a short, sloped plateau. Ours rises over 1100–1500 V and flattens only between 1500 and 1550 V, with a slope of 8.5 ± 0.4% per 100 V.

The rate rises from 772 to 8078 s^{-1} over 1100–1300 V, nearly levels off between 1300 and 1400 V, rises steeply again to 22158 s^{-1} at 1500 V, and levels off a second time. The differential curve (Figure 6b, computed as for the alphas) shows this as two groups of pulses: a smaller peak near 1250 V and a larger one near 1425 V, with a dip at 1350 V.

The shape may look like the counting curve of a mixed alpha and beta source, which has

an alpha plateau followed by a beta plateau (Fig. 1), but it cannot be since Cl-36 is a pure beta emitter. The first plateau more likely comes from a group of large beta pulses, from betas that leave more energy in the gas on long oblique paths or after backscattering from the source backing [2]. A gain change partway through the scan (Section 5.3) could also produce a step, and without the spectrum we cannot rule it out.

The beta pulses are much smaller than the alpha pulses. At the same amplifier gain, alpha pulses cleared a 2.23 V LLD from 400–500 V, while beta pulses needed 1100 V to clear a 0.36 V LLD, and the beta centroid stayed at the LLD up to 1450 V, as expected (Section 2.5).

5.6 Absolute and intrinsic efficiency

We use Eqs. 8 and 9 with $\Omega = 2\pi$, since decay since August 1 is negligible (Eq. 10).

Efficiency

$$r = 1428191/(60\text{ s}) = 23803\text{ s}^{-1}, \qquad \sigma_r = \sqrt{1428191}/(60\text{ s}) = 20\text{ s}^{-1}$$
$$A = 1.032\ \mu\text{Ci} \times 3.7 \times 10^4\ \text{Bq}/\mu\text{Ci} = 38184\text{ Bq}$$
$$\varepsilon_{\text{abs}} = 23803/38184 = 0.6234, \qquad \sigma = 0.6234 \times 20/23803 = 0.0005$$
$$\varepsilon_{\text{int}} = 0.6234 \times 4\pi/2\pi = 1.247, \qquad \sigma = 0.001$$

Section 2.6 predicts $\varepsilon_{\text{abs}} \approx 0.5$ and $\varepsilon_{\text{int}} \approx 100\%$. We measured $\varepsilon_{\text{abs}} = 0.6234 \pm 0.0005$ and $\varepsilon_{\text{int}} = 1.247 \pm 0.001$. An intrinsic efficiency above 100% is not physical, so something added counts or the marked activity understates the source's beta output.

A likely cause is backscattering from the source backing (Section 2.6). If a fraction η of the betas emitted into the backing come back out, the absolute efficiency becomes $0.5(1 + \eta)$, so our result needs $\eta = 2(0.6234) - 1 = 0.25$. Backscatter fractions of this size occur for electrons of a few hundred keV on medium-Z backings such as copper (Section 2.6), so backscatter alone can explain the excess. Two smaller effects push the same way. The count rate was $7.4 \pm 0.2\%$ higher than on the beta counting curve at the same voltage, LLD and gain, which points to the same gain change as in Section 5.3. On a short, sloped plateau, a higher gain brings in more small pulses (Section 2.5).

Background is not a likely cause. We did not take a background count, but even at the $\sim$12 s^{-1} background of our previous labs [4], subtracting it changes ε_{abs} by only 0.0003 (0.05%), far too little to bring it below 0.5. To pin down the cause we should have taken a background count, checked the source certificate, repeated the measurement with the calibrated Am-241 alpha source, which avoids beta backscatter, and extended the beta counting curve past 1550 V so the operating point sits in the middle of a confirmed plateau rather than at its start.

5.7 Gamma response

Gamma rate at 1500 V and plateau slope over 1500–1600 V

$$r_{1500} = 51324/(30\text{ s}) = 1710.8\text{ s}^{-1}, \qquad \sigma = \sqrt{51324}/(30\text{ s}) = 7.6\text{ s}^{-1}$$

$$s_{\%} = 100\,\frac{1712.1 - 1710.8}{1710.8} = 0.1\%\text{ per 100 V}, \quad \sigma_{s_{\%}} = 100\,\frac{\sqrt{7.6^2 + 7.6^2}}{1710.8} = 0.6\%\text{ per 100 V}$$

Section 2.7 predicts a weak response and a broad, beta-like curve. Our Co-60 rate peaks at about 1710 s^{-1}, against about 22000 s^{-1} for Cl-36 betas at the same voltage. The curve in Figure 6a rises from 1200 V and is flat over 1500–1600 V (0.1 ± 0.6% per 100 V), then falls by 16% by 1700 V. The amplifier on the scope showed saturation, so a high gain was likely the cause. Like with the alpha curve, we would repeat the experiment with a lower gain, but did not in this session due to time constraints.

6 Conclusions

The ballistic deficit at 0.5 μs relative to 10 μs was 28.85 ± 0.05%, and we chose a 3 μs shaping time, where the centroid levels off. The alpha and beta counting curves both had the expected shapes, with short, sloped plateaus of 8.4 ± 0.2% and 8.5 ± 0.4% per 100 V, and the beta pulses were much smaller than the alpha pulses. The 3 μs alpha peak gave an energy calibration factor of 2.317 ± 0.001 keV/channel at 800 V and a gain of 50. The absolute efficiency of 0.6234 ± 0.0005 and intrinsic efficiency of 1.247 ± 0.001 are above the physical limits of 0.5 and 100%, most likely because of backscattering from the source backing. The Co-60 gamma rate stayed low as expected, peaking at 1712 ± 8 s^{-1}, while both the alpha and gamma curves dropped at high voltage where the amplifier output saturated. In a future experiment we would read the lab manual more carefully beforehand, lower the amplifier gain before the peak reaches full scale, and check a reference peak at the start of each section to track gain drift. We would also repeat the efficiency measurement with the calibrated Am-241 source and extend the beta counting curve past 1550 V to confirm the plateau.

References

[1] University of Michigan, Department of Nuclear Engineering and Radiological Sciences, "NERS 515 nuclear measurements laboratory: Laboratory 5, proportional counter," 2026.

[2] G. F. Knoll, *Radiation Detection and Measurement*. Hoboken, NJ: John Wiley & Sons, Inc., 4th ed., 2010.

[3] C. Li and C. A. Korfiatis, "NERS 515 laboratory 2: Single-channel and multichannel pulse analysis," 2026. Lab report, University of Michigan.

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A Count rates

HV (V)	Rate (s^{-1})	HV (V)	Rate (s^{-1})
300	0.00 ± 0.00	725	4035.3 ± 11.6
400	0.77 ± 0.16	750	7467.3 ± 15.8
500	13.27 ± 0.66	775	9832.4 ± 18.1
550	83.9 ± 1.7	800	13006.2 ± 20.8
600	372.1 ± 3.5	850	18773.9 ± 25.0
625	665.7 ± 4.7	900	20583.9 ± 26.2
650	1073.0 ± 6.0	1000	22322.5 ± 27.3
675	1552.4 ± 7.2	1050	9536.8 ± 17.8
700	2066.0 ± 8.3		

Table 6: Alpha count rates, $r = N/30$ s, $\sigma_r = \sqrt{N}/30$ s.

HV (V)	Cl-36 beta (s^{-1})	Co-60 gamma (s^{-1})
1000	0.0 ± 0.0	0.0 ± 0.0
1100	772.0 ± 5.1	–
1200	3260.3 ± 10.4	349.7 ± 3.4
1300	8077.5 ± 16.4	825.9 ± 5.2
1400	11512.4 ± 19.6	1457.0 ± 7.0
1450	17394.2 ± 24.1	–
1500	22157.5 ± 27.2	1710.8 ± 7.6
1550	23096.1 ± 27.7	1688.3 ± 7.5
1600	–	1712.1 ± 7.6
1650	–	1621.1 ± 7.4
1700	–	1434.7 ± 6.9

Table 7: Beta and gamma count rates, $r = N/30$ s, $\sigma_r = \sqrt{N}/30$ s.